

Electronic and Magnetic Structure of FeS₂ Marcasite

Ivan Lopez Mata (77763A), Randy Wendel Mojica Flores (82234A), Manahil Noor (81176A)

Hard Matter: Fundamentals and Applications

Università degli Studi di Milano, V. Golgi 19, 20133 Milan, Italy

Prof. Rocco Martinazzo and Prof. Davide Ceresoli

July 22, 2026

Abstract

Marcasite is a metastable transition-metal dichalcogenide that coexists in nature with the thermodynamically more stable cubic pyrite polymorph. We investigate the electronic and magnetic properties of the orthorhombic marcasite polymorph of FeS₂, using implementations within Kohn–Sham DFT at the GGA–PBE functional parametrization level. All calculations converge to a non-magnetic ground state ($S = 0$), regardless of the initial ferromagnetic configuration, confirming a low-spin Fe²⁺ configuration consistent with experimental temperature-independent diamagnetism. While ultrasoft-pseudopotential results suggest a marginal semimetal, *GW*-tuned PAW pseudopotentials predict a conventional indirect gap of 0.849 eV, aligning closely with the experimental optical gap of 0.76 eV reported for high-purity single crystals. Analysis of the projected density-of-states reveals that band-edge states are consistently derived from hybridized Fe *3d*–S *3p* manifolds rather than purely atomic orbitals. These findings indicate that while marcasite’s non-magnetic nature is a robust result, recovering its narrow-gap semiconducting character requires high-accuracy treatment of core and semicore electrons.

1 Introduction

Iron disulfide, FeS₂, is a first row transition-metal dichalcogenide that occurs naturally in the mineral polymorphs marcasite and pyrite, the latter also known as “fool’s gold” [1, 2]. It has attracted sustained academic interest over the last decades due to its wide natural abundance, low cost, non-toxicity under standard conditions of pressure and temperature, strong visible-light absorption, and favorable electrochemical properties [3]. For these reasons, it is extensively discussed in the literature for potential applications in photovoltaics, photodetectors, photocatalysis, and metal-ion battery electrodes [1].

From the structural point of view, the pyrite and marcasite phases coexist naturally and are generally described as non-metallic materials. The marcasite phase crystallizes in an orthorhombic system with space group $Pn\bar{n}m$ (No. 58) and is thermodynamically metastable with respect to the pyrite phase, which crystallizes in the cubic space group $Pa\bar{3}$ [1, 4]. Marcasite is less common than pyrite in nature, being primarily found in acidic hydrothermal environments, and when exposed to high temperatures or air during extended periods of time, it transforms into pyrite [4, 2]. While pyrite can be synthesized without much complications in pure form, obtaining pure marcasite had been more challenging because of frequent intergrowth of the two phases. Recently, improvements of synthetic conditions allowed the formation of pure marcasite single crystals free of the sulfur vacancies that plague natural samples. This enabled a reliable experimental measurement of marcasite’s electronic properties: an allowed direct optical band gap of 0.76 eV, an electronic transport gap of 0.11 eV, temperature-independent diamagnetism, and a soft X-ray absorption/emission (XAS/XES) band gap of 0.73 eV, value remarkably close to pyrite, 0.87 eV, obtained with the same experimental technique [2].

From the magnetic side, previous studies suggest a very weak and almost temperature-independent magnetic response, more consistent with a low-spin Fe state exhibiting Van Vleck paramagnetism, which is temperature-independent diamagnetism, rather than true ferromagnetism [5, 6]. Recent measurements on high-purity single crystals confirm this behavior, consistent with a fully-paired $3d^6$ low-spin configuration at the Fe^{2+} center [4, 2].

It is also useful to place marcasite within the broader family of isostructural marcasite-type compounds TMX_2 , where TM corresponds to a transition metal, and X to a chalcogen or pnictogen. The characteristic edge-sharing TMX_6 octahedral connectivity along the crystallographic c -axis, common to the whole marcasite family, tends to render specific $3d$ orbitals (e.g., $3d_{zx}$, $3d_{zy}$) essentially non-bonding, which in several members of this family, such as FeSb_2 , CrSb_2 , and MnSb_2 , gives rise to remarkably flat electronic bands near the Fermi level [7]. This provides useful physical context for interpreting the localized, weakly dispersive valence-band features that we also observe in FeS_2 marcasite below.

Henceforth, we performed first principles electronic structure calculations of FeS_2 , focused in the polymorph marcasite, in order to characterize its structure while exploring its density of states at the Fermi surface and its band gap, as well as confirming its magnetic behavior after the self-consistent cycles.

2 Theory and Method

2.1 Structural Model

To describe marcasite, stoichiometrically FeS_2 , a naive electron-counting approach might suggest magnetic behavior due to the partially filled Fe $3d$ shell (Fe: $[\text{Ar}] 3d^6 4s^2$). However, the bonding in marcasite is more accurately described through persulfide formation: $\text{Fe}^{2+}[\text{S}_2]^{2-}$ [1].

In this model, iron is formally divalent, Fe^{2+} , while sulfur exists as persulfide dimers, S_2^{2-} , with an S–S bond order of approximately one. The formation of these S–S dimers is a critical structural feature that significantly influences the electronic structure [1, 4]. The S_2^{2-} unit accepts electrons from Fe, stabilizing the $3d^6$ configuration. Significant Fe $3d$ –S $3p$ hybridization modifies the crystal-field splitting, and the S–S bonds impose geometric constraints that influence the Fe coordination environment.

The $Pn\bar{n}m$ orthorhombic unit cell of marcasite contains 6 atoms: 2 Fe atoms and 4 S atoms, with iron atoms arranged in the body centered cubic sites (Figure 1). The sulfur atoms form S–S dimers along the c -axis, a defining feature of the marcasite structure [4]. Unlike pyrite, where all S–S dimers are equivalent and aligned along the cubic body diagonals, marcasite exhibits two distinct S–S dimer orientations within the unit cell due to the lower symmetry. Each Fe atom is coordinated by six sulfur atoms forming a distorted FeS_6 octahedron, with a point group approximately D_{2h} rather than O_h , which corresponds to the building block of both marcasite and pyrite, both phases differing mainly in the way the edges and vertex connect to each other. This distortion in FeS_2 lifts the degeneracy of the t_{2g} and e_g orbitals, modifies the crystal-field splitting parameter Δ_{oct} , and influences magnetic exchange interactions. The initial crystallographic information used in the calculations is summarized in Table 1.

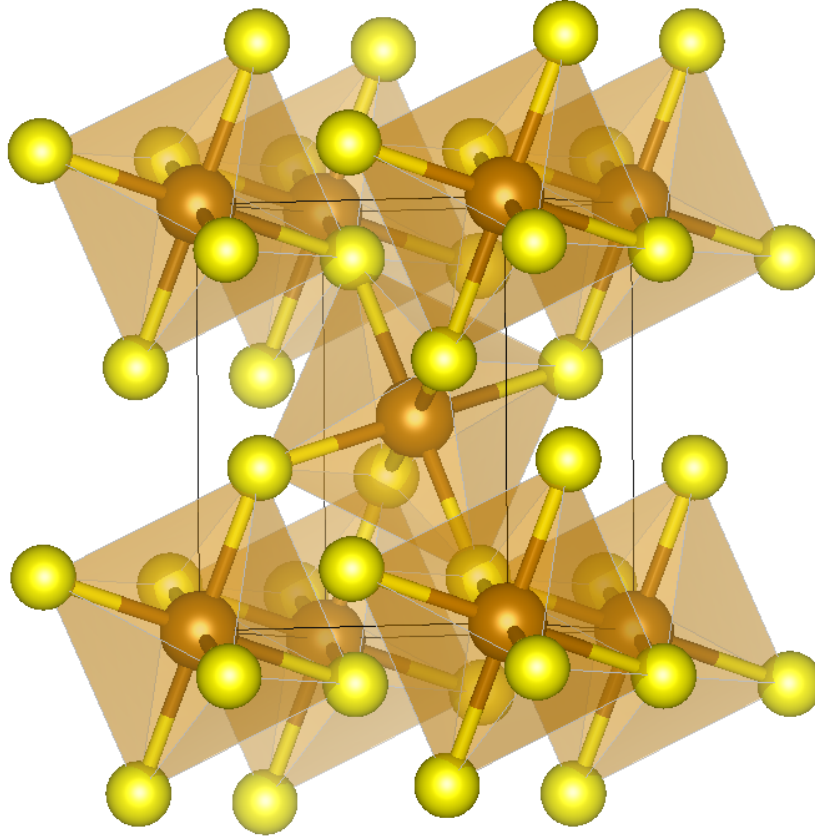


Figure 1: Crystal structure of the FeS_2 polymorph, marcasite.

Table 1: Initial crystallographic and computational setup for marcasite FeS_2 .

Quantity/Item	Value/Description
Space group	$Pn\bar{m}m$ (No. 58)
Atoms per cell	6 (2 Fe + 4 S)
Valence electrons per cell	40 (Eq. (1))
Lattice parameters ($\text{\AA}$)	$a = 3.37770$, $b = 4.42562$, $c = 5.39653$
Cell volume ($\text{\AA}^3$)	$V = 80.6696$
Initial SCF mesh (k -points)	$7 \times 6 \times 4$
Final magnetic state	Self-consistent non-magnetic ($S = 0$)

2.2 Composition, Valence Electron Count, and Formal Oxidation States

The primitive marcasite cell contains 6 atoms (2 Fe + 4 S). Using the pseudopotential valence-electron count relevant to the Perdew–Burke–Ernzerhof (PBE) calculations [8], each Fe atom contributes its $3d^6 4s^2$ shell (8 valence electrons), and each S atom contributes its $3s^2 3p^4$ shell (6 valence electrons), giving a total of

$$N_{\text{val}} = 2 \times 8 + 4 \times 6 = 40 \text{ valence electrons per unit cell.} \quad (1)$$

Assigning formal oxidation states [9] through the persulfide model $\text{Fe}^{2+}[\text{S}_2]^{2-}$, each Fe^{2+} center has configuration $[\text{Ar}]3d^6 4s^0$, while each S^- in the S–S dimer has configuration $[\text{Ne}]3s^2 3p^5$. In principle, the $3d^6$ configuration of Fe^{2+} can be either low-spin ($t_{2g}^6 e_g^0$, $S = 0$) or high-spin

($t_{2g}^4 e_g^2$, $S = 2$) depending on the relative magnitude of the octahedral crystal-field splitting Δ_{oct} and the electron pairing energy P . According to theory, an open shell configuration is possible, which means that the system could be magnetic. However, a closed shell electron count suggests that electrons are paired (low spin and high crystal field energy), in other words, marcasite should behave as a non-magnetic, narrow-gap semiconductor rather than a metal.

2.3 Computational Method

The first principles electronic-structure calculations for FeS₂, are performed using Kohn–Sham (KS) Density Functional Theory (DFT) with the plane waves basis set implemented in QUANTUM ESPRESSO (QE) [10, 11]. The many-electron problem is mapped onto a fictitious system of non-interacting electrons moving in an effective potential obtained by solving the self-consistent-field (SCF) cycle of the KS equations [12, 13]:

$$H^{KS}\psi_i(\mathbf{r}) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{ext}}(\mathbf{r}) + V_{\text{Har}}(\mathbf{r}) + V_{\text{xc}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r}), \quad (2)$$

$$V_{\text{Har}}(\mathbf{r}) = \frac{\delta E_{\text{Har}}}{\delta n(\mathbf{r})} = e^2 \int d\mathbf{r}' \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \quad n(\mathbf{r}) = \sum_{i=1}^N \psi_i^*(\mathbf{r})\psi_i(\mathbf{r}), \quad (3)$$

$$V_{\text{xc}}(\mathbf{r}) = \frac{\delta E_{\text{xc}}}{\delta n(\mathbf{r})}, \quad (4)$$

where $V_{\text{ext}}(\mathbf{r})$ the external potential of the nuclei, $V_{\text{Har}}(\mathbf{r})$ the Hartree potential, $n(\mathbf{r})$ is the electron density, N the orbital occupations, and $V_{\text{xc}}(\mathbf{r})$ the exchange-correlation potential. The E_{xc} part of the Hamiltonian is evaluated at the Generalized Gradient Approximation (GGA) level with the parametrization of the Perdew–Burke–Ernzerhof (PBE) functional (Eq. (5)) [8],

$$E_{\text{xc}}^{\text{PBE}}(n) = \int d\mathbf{r} n(\mathbf{r}) \varepsilon_{\text{xc}}^{\text{PBE}}(n(\mathbf{r}), |\nabla n(\mathbf{r})|), \quad (5)$$

which improves on the local density approximation by making the exchange-correlation energy density depend explicitly on the local density gradient, at moderate additional computational cost.

We tested three different pseudopotentials to check the robustness of our results. At first, we used a standalone ultrasoft pseudopotential with the GGA–PBE functional and nonlinear core correction for calculations with a converged kinetic energy (ecutwfc) and charge density (ecutrho) cutoff of 35 and 350 Ry for the SCF cycles. The electronic configuration of the valence shell for the involved atom types were $3s^2 3p^6 3d^6 4s^2 4p^0$ for Fe (55.845 a.m.u.), and $3s^2 3p^4$ for S (32.06 a.m.u.).

Additional confirmation calculations were performed to observe if there was an improvement of the description of the Fe atoms in FeS₂ marcasite, using a Rappe Rabe Kaxiras Joannopoulos ultrasoft pseudopotential (RRKJUS) with the GGA–PBE functional, which focuses in improving the convergence characteristics of the plane-wave basis [14], obtained from pslibrary (v.1.0.0) [15]. The electronic configuration for the Fe valence electrons are described as $3s^2 4s^2 3p^6 3d^6$ with core corrections, using ecutwfc and ecutrho of 71 and 710 Ry, as recommended for this pseudopotential. This pseudopotential was reported to reproduce the band structure of the analogous system pyrite [16].

A final comparison calculation was made using the Vienna ab initio Simulation Package (VASP) with the projector augmented wave method (PAW) [17, 18] and the GW-tuned approximately norm-conserving PAW pseudopotentials [19], which treats explicitly the 3s, 3p semi-core states in the valence part for Fe in the exchange-correlation functional, with a valence electronic configuration for Fe described as $3s^2 3p^6 3d^7 4s^1$, using a cutoff for the plane-wave basis set of 293.238 eV (≈ 21.5 Ry). This was reported to reproduce with great accuracy the experimental

band gap and the characteristics of the KS states near the Fermi level for the FeS₂ polymorphs marcasite and pyrite, as well as the stability of their phases [1].

The Brillouin zone for FeS₂(*Pnnm*) was sampled using a Γ -centered $7 \times 6 \times 4$ Monkhorst-Pack k -point mesh (Figure 2) [20]. To optimize the atomic coordinates, structural variable-cell optimization was carried out via the `vc-relax` algorithm implemented in QUANTUM ESPRESSO, starting from the experimental/primitive-cell coordinates. After relaxation, we performed SCF to get ground state electron density and denser NSCF calculations for Density of States (DOS), fermi surface and band-structure analysis. Semicore states were discarded from the electronic plots to focus on the Fermi-level region.

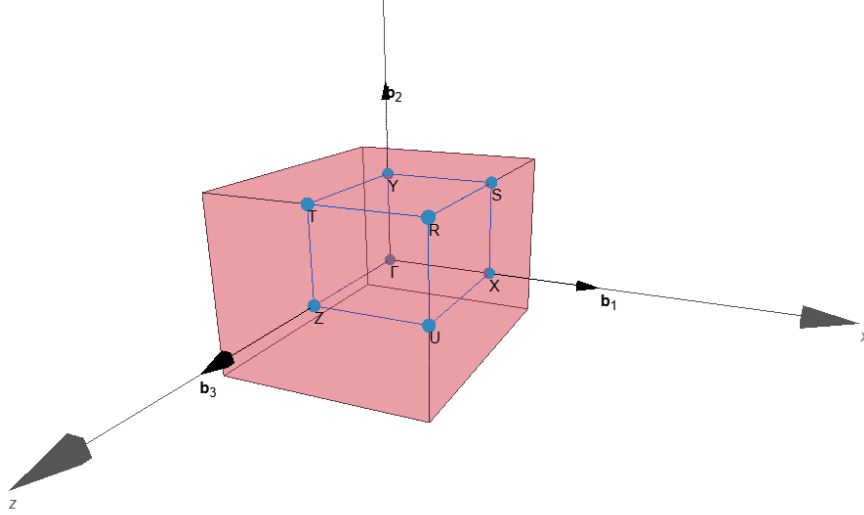


Figure 2: BZ k -path for the FeS₂ polymorph, marcasite.

2.4 Ligand-Field Analysis and Magnetism

In the distorted octahedral environment, the Fe 3*d* manifold splits into a t_{2g} triplet and an e_g doublet separated by the crystal-field splitting energy Δ_{oct} . For a $3d^6$ ion, the ground-state spin configuration is determined by the competition between Δ_{oct} and the electron pairing energy P :

$$\text{low-spin: } (t_{2g}^6 e_g^0, S = 0) \text{ if } \Delta_{\text{oct}} > P, \quad \text{high-spin: } (t_{2g}^4 e_g^2, S = 2) \text{ if } \Delta_{\text{oct}} < P. \quad (6)$$

In marcasite FeS₂, a strong Fe–S covalent character and back-donation from the π - into the σ_p^* orbital in S_2^{2-} , push the e_g antibonding states upward in energy, favoring the low-spin regime, i.e. $\Delta_{\text{oct}} > P$, and hence a fully paired, non-magnetic t_{2g}^6 configuration [1].

The same physical principle can be recast through the Stoner criterion for band ferromagnetism,

$$I \cdot N(E_F) > 1, \quad (7)$$

where I is the Stoner exchange parameter and $N(E_F)$ the density of states (DOS) at the Fermi level. When electrons are all paired up in a low-spin configuration, there are no electrons available at the Fermi level to align magnetically. According to the Stoner criterion, there should be present a huge amount of electrons at the Fermi level for the material to display magnetic properties. Since the literature has reported that FeS₂ has a gap at E_F , meaning that no electrons are present at that energy level, this material cannot be magnetic. To confirm this, we carried out a calculation starting with a magnetic guess. We treated spin-up and spin-down electrons separately, giving them different energies with the local magnetization as $m(\mathbf{r}) = n_{\uparrow}(\mathbf{r}) - n_{\downarrow}(\mathbf{r})$. Then, after letting the calculation find the lowest energy state, the

magnetic guess collapsed into a non-magnetic regime during the calculation, proving that the non-magnetic state corresponds to the true ground state.

3 Results and Discussion

3.1 Structural and Magnetic Relaxation

The converged ground state was found to be non-magnetic, consistent with the low-spin scenario of Eq. (6) and with the Stoner-criterion argument of Eq. (7). Although the three calculations were performed with different pseudopotential implementations and relaxation algorithms, all of them evolve toward a stationary structural solution (Figure 3).

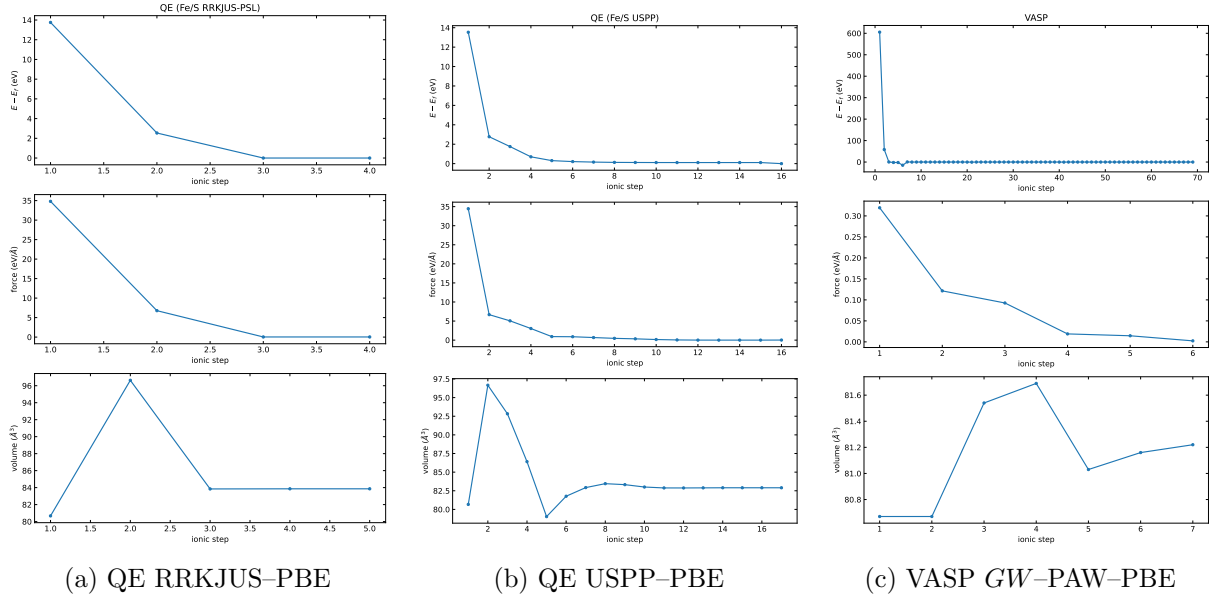


Figure 3: Structural-relaxation histories obtained with the two Quantum ESPRESSO ultrasoft pseudopotential sets a) RRKJUS-PBE, b) USPP-PBE, and c) GW-PAW-PBE with VASP.

The relaxation paths should be compared qualitatively rather than point by point. In particular, each energy curve is referred to its own final energy, and therefore the initial energy differences visible in Figure 3 are not absolute energy differences between methods. The first QE calculation reaches its final plateau in only a few reported ionic steps, whereas the QE-USPP-PBE calculation follows a longer and more gradually damped trajectory. The GW-PAW VASP relaxation displays a particularly large initial energy decrease, followed by an apparently smoother evolution of the forces and cell volume. However, the volume curves should not be compared visually without considering the different vertical scales used in the plots. In the VASP calculation, the displayed volume interval is narrower than approximately 1 Å³, whereas the two Quantum ESPRESSO plots span volume variations of nearly 15 Å³. Consequently, the changes in the VASP volume may appear more pronounced within the plotted window, even though their absolute magnitude is considerably smaller. The different relaxation trajectories are therefore more reasonably attributed to the optimizer, starting geometry, basis representation, pseudopotential, and convergence settings, rather than to convergence toward different physical phases.

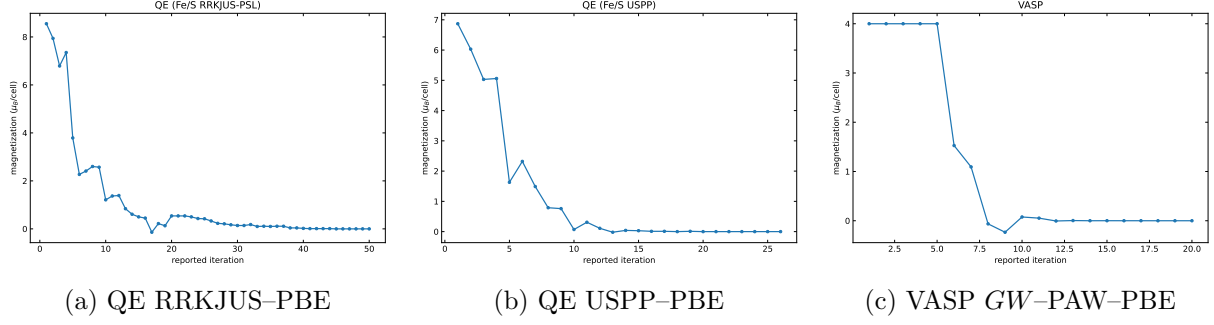


Figure 4: Collapse of the cell magnetization during electronic SCF cycles.

The magnetization histories provide a direct and method-independent indication that the imposed ferromagnetic starting configuration is not stable. The calculations start with finite moments of approximately 4–9 μ_B per cell, depending on the initialization, but the moments decrease progressively and ultimately fluctuate around zero (Figure 4). The small negative or positive values appearing close to convergence are numerical oscillations produced by charge and spin-density mixing; they do not represent a stable antiferromagnetic or ferrimagnetic solution.

The converged state is therefore compatible with a low-spin Fe^{2+} configuration. Strong Fe–S covalency raises the predominantly e_g antibonding states relative to the occupied t_{2g} type of manifold, favouring the condition $\Delta_{\text{oct}} > P$ and hence an $S = 0$ solution.

3.2 Electronic Characterization

The complete band structures are first compared over a broad energy window. This representation is useful for identifying the principal groups of bands and determining whether the three electronic-structure descriptions preserve the same general chemical picture.

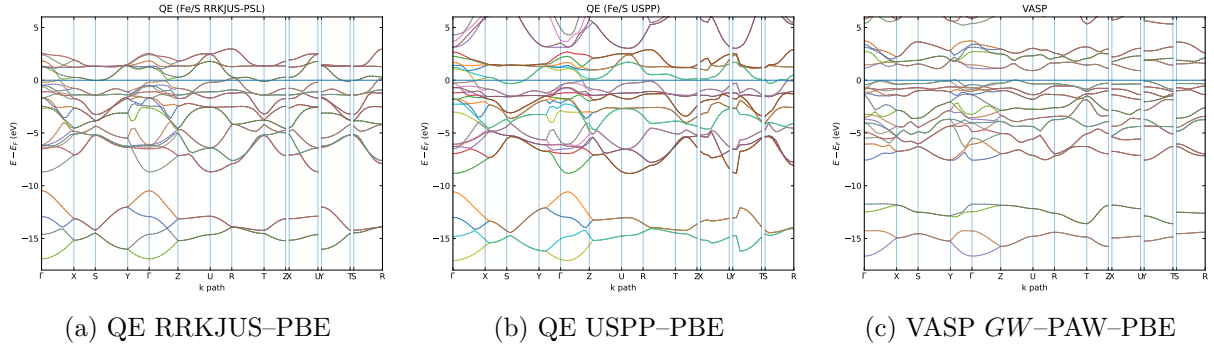


Figure 5: Full spin-polarized band structures of marcasite FeS_2 , aligned with respect to the calculated Fermi level.

At the scale of Figure 5, the three methods produce a broadly comparable grouping of states. Deep, relatively isolated bands appear at approximately -12 to -17 eV, while a denser set of hybridized Fe–S bands occurs closer to the Fermi level. However, the agreement is not exact: individual bands shift, change curvature and occasionally change their relative ordering. These differences become particularly important near the valence- and conduction-band edges.

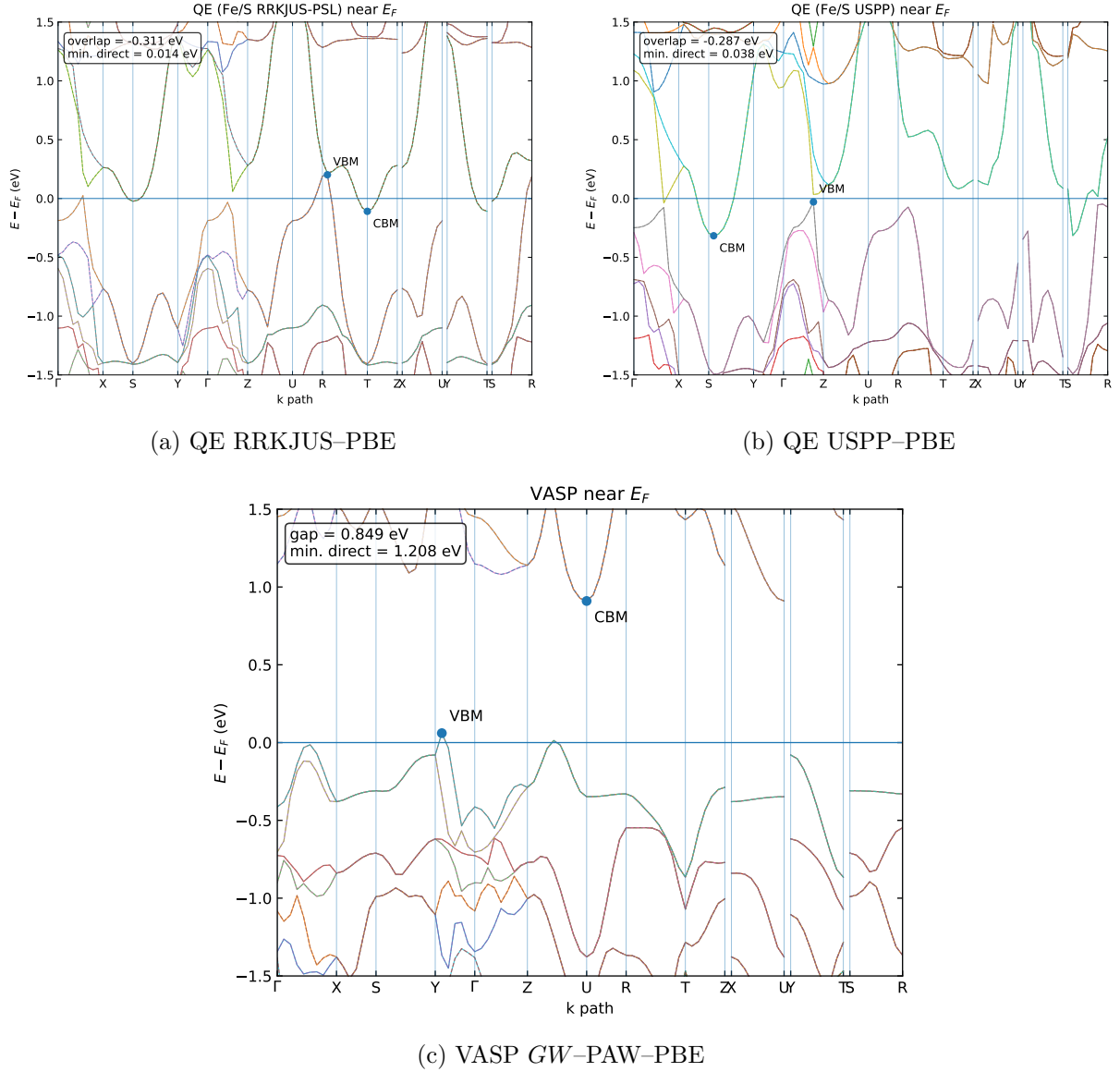


Figure 6: Enlarged view of the states close to the Fermi level portraying the conduction band minimum (CBM) and valence band maximum (VBM), as well as the band gap calculated with the different pseudopotentials explored.

The band-edge analysis reveals a stronger method dependence than is visible in the complete band structures. Using the convention

$$E_g^{\text{ind}} = E_{\text{CBM}} - E_{\text{VBM}}, \quad (8)$$

a negative value of E_g^{ind} indicates that the conduction and valence-band manifolds overlap at different points of the Brillouin zone (Figure 6). It therefore describes a semimetallic solution rather than a true indirect semiconductor for the calculations made with the ultrasoft pseudopotentials, in contrast with the GW -PAW-based pseudopotential.

Table 2: Band-edge quantities extracted from Figure 6. Negative indirect values denote valence–conduction overlap.

Method	E_g^{ind} or overlap (eV)	Minimum direct separation (eV)
QE RRKJUS–PBE	−0.311	0.014
QE USPP–PBE	−0.287	0.038
VASP <i>GW</i> –PAW–PBE	0.849	1.208

Consequently, the three PBE calculations do not support a single, code-independent value for the band gap. Both QE calculations retain a considerably small positive and direct separation at every individual k-point, but the VBM and CBM occur at different k-points and overlap in energy. The VASP result instead predicts a conventional indirect semiconductor with a gap close to 0.85 eV, in agreement with previous experimental and theoretical reports [2, 1].

This comparison evidences that the electronic classification of marcasite FeS₂ is highly sensitive to the potential used to describe the Fe and S cores, as well as to the basis and implementation.

3.3 Total Density of States

The total density of states (DOS) provides an independent representation of the electronic structure and makes the spin symmetry of the converged solutions particularly clear.

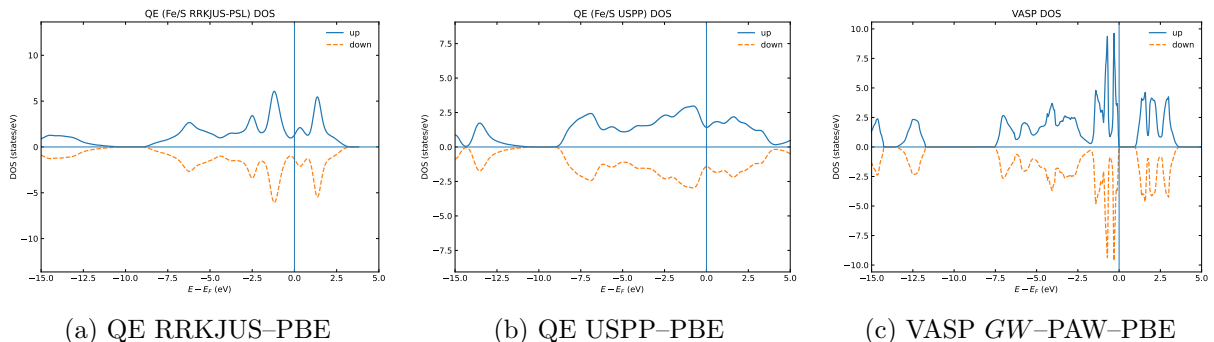


Figure 7: Spin-resolved total density of states for the different pseudopotentials employed: a) RRKJUS–PBE, b) USPP–PBE, and c) *GW*–PAW–PBE. The spin-down component is plotted with a negative sign for visual comparison.

The reflection symmetry between the positive spin-up DOS and the negative spin-down DOS (Figure 7), is consistent with the magnetic convergence shown in Figure 4. This symmetry confirms that the final electronic structures are effectively spin degenerate.

It is important, however, not to confuse spin degeneracy with the presence of an electronic gap. The two QE calculations retain visible spectral weight close to E_F , which is consistent with the negative indirect overlap identified in Figure 6. Part of the intensity may additionally arise from the smearing and DOS broadening employed during post-processing. The VASP result displays a clearer depletion around E_F , in agreement with its positive indirect gap.

Accordingly, the Stoner argument should be applied separately to each calculation. All three datasets converge to zero magnetization, but the reason should not be described universally as a strictly vanishing $N(E_F)$. In the QE calculations, a small or broadened DOS may remain at the Fermi energy without being sufficiently large to stabilize an itinerant ferromagnetic state.

3.4 Projected Density of States

The orbital-projected density of states (pDOS) was used to distinguish the contributions of Fe $3d$ and S $3p$ orbitals. The two QE calculations are shown together because they employ the same electronic-structure code but different pseudopotential constructions.

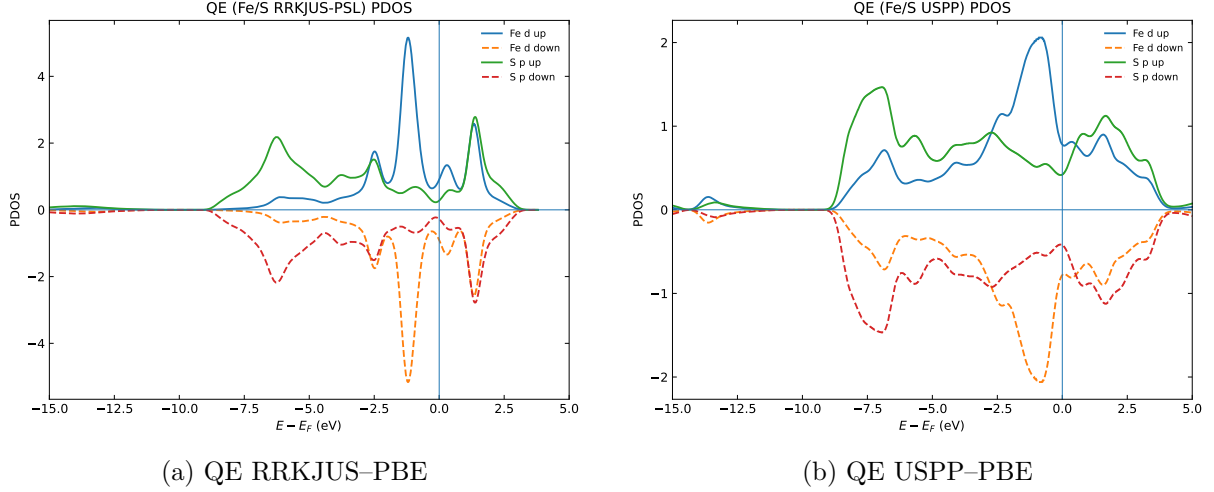


Figure 8: Comparison of the Fe $3d$ and S $3p$ pDOS obtained with the two QE pseudopotential sets a) RRKJUS-PBE and b) USPP-PBE.

In both QE calculations, the states closest to the Fermi energy contain a strong Fe $3d$ contribution mixed with a non-negligible S $3p$ component (Figure 8). The simultaneous occurrence of Fe and S intensity over the same energy intervals demonstrates that the relevant bands cannot be interpreted as purely atomic Fe states. Instead, they originate from covalent Fe-S combinations.

The two pseudopotentials nevertheless distribute the spectral weight differently. The RRKJUS-PBE calculation produces sharper Fe $3d$ features, whereas the USPP-PBE result is more broadly distributed over the upper valence region. Such differences explain why the corresponding band edges and indirect overlaps are not numerically identical.

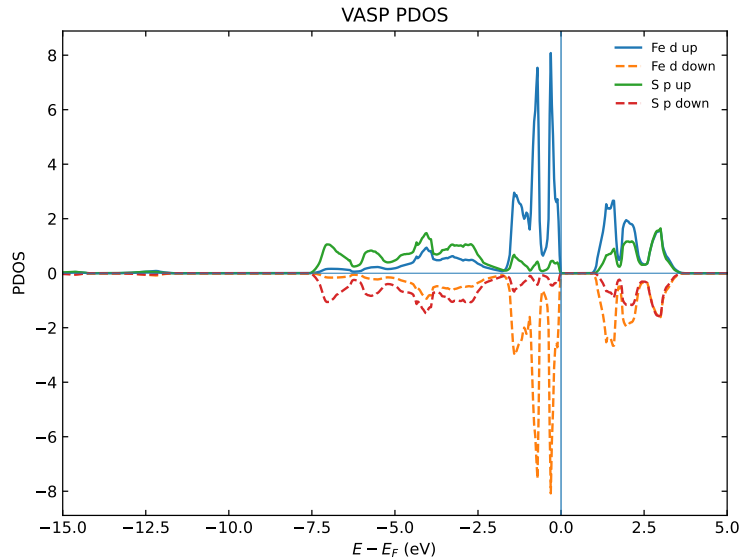


Figure 9: Fe $3d$ and S $3p$ pDOS obtained using VASP with the pseudopotential GW-PAW-PBE.

The VASP projections contain narrower and more intense Fe $3d$ peaks near the upper valence-

band region than the corresponding QE projections (Figure 9). Nevertheless, the qualitative interpretation remains similar: the band-edge states are principally transition-metal d states modified by covalent mixing with ligand p orbitals.

The low-spin interpretation is therefore better expressed in terms of hybridized manifolds derived from crystal-field, rather than perfectly localized atomic orbitals. The occupied band-edge states may be described as predominantly t_{2g} , whereas the lowest unoccupied states contain stronger antibonding character and may be associated predominantly with the manifold derived from e_g .

3.5 Combined Band-Structure and DOS Representations

The combined plots provide a direct visual alignment between the band energies and the corresponding DOS. Although they contain information already discussed separately, they are useful as an internal consistency check because the band extrema and the DOS depletion or accumulation can be examined on the same energy scale.

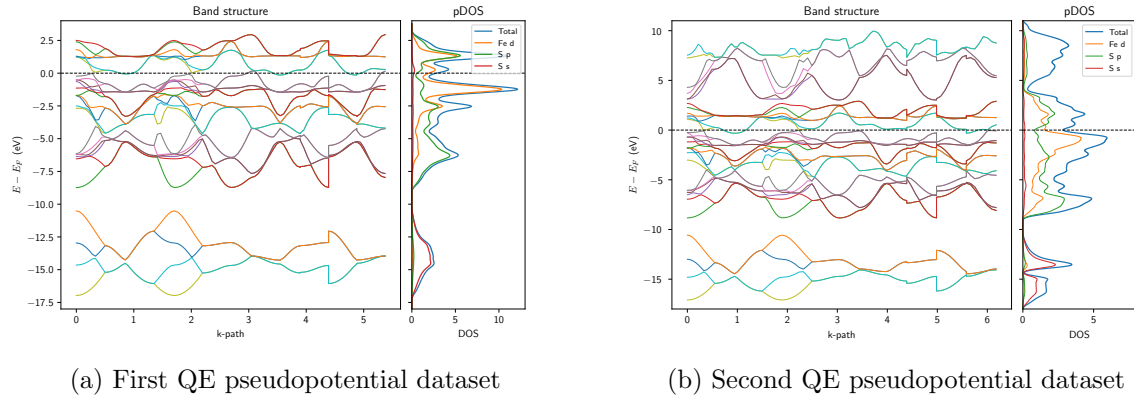


Figure 10: Combined band-structure and DOS representations with the two QE pseudopotentials a) RRKJUS-PBE and b) USPP-PBE.

The joint QE figures make clear why a small positive direct separation does not necessarily produce a zero DOS at the Fermi energy (Figure 10). Bands may remain separated at every fixed $\mathbf{k}$ while still overlapping indirectly when their extrema occur at different positions along the Brillouin-zone path. Smearing and interpolation can then produce additional finite DOS intensity around E_F .

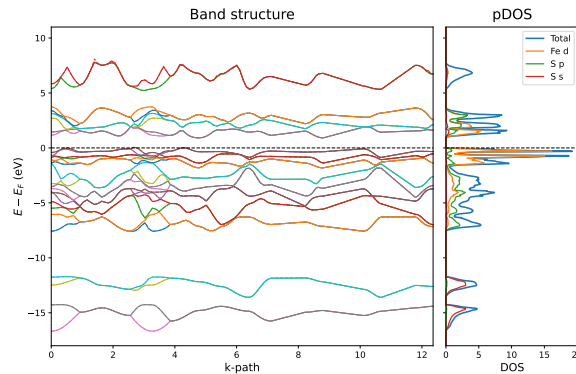


Figure 11: Combined band-structure and DOS representation obtained with VASP pseudopotential.

In the VASP result, the band structure and DOS provide a mutually consistent semiconducting picture (Figure 11). The band extrema remain separated in energy, and the corresponding DOS is depleted in the same interval. This agreement contrasts with the QE solutions and emphasizes that the predicted gap is not determined by the PBE functional alone, but also by the inner description of the terms in the pseudopotential, basis representation, relaxation and numerical settings.

The present calculations nevertheless show that the precise electronic classification of FeS₂ is numerically delicate. Depending on the potential and implementation, the PBE functional produces either a small indirect band overlap or a finite semiconducting gap. The most robust conclusion common to all three calculations is therefore the non-magnetic character of the converged state and the strong Fe 3*d*–S 3*p* hybridization, rather than a unique numerical value of the band gap, as these inconsistencies in the calculation of the electronic properties and stability of FeS₂ have been widely reported in the literature [1, 21, 22, 23].

Nonetheless, the main limitation of the present study lies in the execution of similar first principles calculation procedures for computing the marcasite electronic properties in FeS₂, using different software packages, which leads to the introduction of variables pertaining to internal functioning of the *ab initio* software, and diminishing the meaning of the results obtained. However, this results are useful from a qualitative point of view to address the main difference between the pseudopotentials and softwares used in order to find which provides a better electronic description of the system of interest, in this case marcasite FeS₂, allowing to explore more in depth a specific type of functional to gain a deeper insight into the microscopic particularities of the system, while reproducing with more accuracy its experimental properties, as was described by Zhang et al in its testing of several pseudopotentials and functionals available in the VASP package in order to elucidate the relative stability between both FeS₂ polymorphs competing phases, marcasite and pyrite [1].

4 Conclusions

In the present study, KS-DFT was used to study the orthorhombic marcasite phase of FeS₂. Independent DFT-PBE calculations confirm that orthorhombic marcasite FeS₂ is non-magnetic in its ground state ($S = 0$). In every case, initial ferromagnetic moments collapsed during self-consistency, consistent with a low-spin Fe²⁺ (3*d*⁶) configuration. This behavior is driven by strong Fe–S covalency and a crystal field splitting where $\Delta_{\text{oct}} > P$, leaving no partially occupied states to sustain itinerant magnetism via the Stoner mechanism.

The three calculations diverge noticeably in their description of the band gap. While the ultrasoft pseudopotential implementations in QUANTUM ESPRESSO suggest a marginal semimetal with a small negative energetic overlap, the *GW*-tuned PAW pseudopotential in VASP recovers a proper indirect gap of 0.849 eV, which aligns closely with experimental optical gap of 0.76 eV. This discrepancy highlights the sensitivity of FeS₂ to the treatment of Fe semicore states and the pseudopotential construction.

Projected density of states (pDOS) analysis across all methods consistently shows that band-edge states are mainly hybridized Fe 3*d*–S 3*p* combinations. This supports the persulfide bonding model and the t_{2g}/e_g splitting used to explain the low-spin state. These findings mirror reported difficulties in predicting the relative stability of FeS₂ polymorphs, where semilocal functionals often fail to resolve the energetic hierarchy between marcasite and pyrite. The study’s primary limitation involves disentangling genuine physical trends from cross-code variables such as core treatment and cutoff convergence. Nonetheless, the comparison identifies pseudopotential choice as the main driver of band gap variance. Future work could employ hybrid functionals within a unified framework to preserve the robust non-magnetic state while matching experimental optical properties with high precision.

References

- [1] M.-Y. Zhang, Z.-H. Cui, and H. Jiang, Realative stability of FeS_2 polymorphs with the random phase approximation approach, *Journal of Materials Chemistry A* **6** (2018) 6606–6616.
- [2] K. Ma, R. Lefèvre, Q. Li, J. Lago, O. Blacque, W. Yang, and F. O. von Rohr, Synthetic control over polymorph formation in the d-band semiconductor system FeS_2 , *Chemical Science* **12** (2021) 13870.
- [3] H. Qin, J. Jia, L. Lin, H. Ni, M. Wang, and L. Meng, Pyrite FeS_2 nanostructures: Synthesis, properties and applications, *Materials Science and Engineering B* **236–237** (2018) 104–124.
- [4] X. Li, S.-H. Qi, W.-C. Zhang, Y.-Z. Feng, and J.-M. Ma, Recent progress on FeS_2 as anodes for metal-ion batteries, *Rare Metals* **39** (2020) 1239–1255.
- [5] S. Miyahara and T. Teranishi, Magnetic Properties of FeS_2 and CoS_2 , *Journal of Applied Physics* **39** (1968) 896–898.
- [6] M.S. Seehra and M.S. Jagadeesh, Temperature-dependent magnetic susceptibility of marcasite (FeS_2), *Physical Review B* **20** (1979) 3897–3899.
- [7] C.J. Linnemann, K.-K. Huynh, D. Ceresoli, and M. Bremholm, Flat band in multi-band metal MnSb_2 , *arXiv* (2026) 2604.25324.
- [8] J.P. Perdew, K. Burke, and M. Ernzerhof, Generalized Gradient Approximation Made Simple, *Physical Review Letters* **77** (1996) 3865.
- [9] A. Walsh, A.A. Sokol, J. Buckeridge, D.O. Scanlon, and C. R.A. Catlow, Oxidation states and ionicity, *Nature Materials* **17** (2018) 958–964.
- [10] P. Giannozzi, S. Baroni, N. Bonini, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, G.L. Chiarotti, M. Cococcioni, I. Dabo, A. Dal Corso, S. de Gironcoli, S. Fabris, G. Fratesi, R. Gebauer, U. Gerstmann, C. Gougoussis, A. Kokalj, M. Lazzeri, L. Martin-Samos, N. Marzari, F. Mauri, R. Mazzarello, S. Paolini, A. Pasquarello, L. Paulatto, C. Sbraccia, S. Scandolo, G. Sclauzero, A.P. Seitsonen, A. Smogunov, P. Umari, and R.M. Wentzcovitch, QUANTUM ESPRESSO: a modular and open-source software project for quantum simulations of materials, *Journal of Physics: Condensed Matter* **21** (2009) 395502.
- [11] P. Giannozzi, O. Andreussi, T. Brumme, O. Bunau, M. Buongiorno Nardelli, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, M. Cococcioni, N. Colonna, I. Carnimeo, A. Dal Corso, S. de Gironcoli, P. Delugas, R.A. DiStasio Jr., A. Ferretti, A. Floris, G. Fratesi, G. Fugallo, R. Gebauer, U. Gerstmann, F. Giustino, T. Gorni, J. Jia, M. Kawamura, H.-Y. Ko, A. Kokalj, E. Küçükbenli, M. Lazzeri, M. Marsili, N. Marzari, F. Mauri, N.L. Nguyen, H.-V. Nguyen, A. Otero-de-la-Roza, L. Paulatto, S. Poncé, D. Rocca, R. Sabatini, B. Santra, M. Schlipf, A.P. Seitsonen, A. Smogunov, I. Timrov, T. Thonhauser, P. Umari, N. Vast, X. Wu, and S. Baroni, Advanced capabilities for materials modelling with QUANTUM ESPRESSO, *Journal of Physics: Condensed Matter* **29** (2017) 465901.
- [12] W. Kohn and L.J. Sham, Self-Consistent Equations Including Exchange and Correlation Effects, *Physical Review* **140** (1965) A1133.
- [13] R.M. Dreizler and E.K.U. Gross, Density Functional Theory: An Approach to the Quantum Many-Body Problem, *Springer-Verlag, Berlin* (1990).
- [14] A.M. Rappe, K.M. Rabe, E. Kaxiras, and J.D. Joannopoulos, Optimized pseudopotentials, *Physical Review B* **41** (1990) 13175.

- [15] A. Dal Corso, Pseudopotentials periodic table: From H to Pu, *Computational Materials Science* **95** (2014) 337-350.
- [16] A. Mukherjee and A. Subedi, Impact of hole-doping on the thermoelectric properties of pyrite FeS₂, *Solid State Communications* **406** (2025) 116194.
- [17] G. Kresse and J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, *Physical Review B* **54** (1996) 11169.
- [18] G. Kresse and D. Joubert, From ultrasoft pseudopotentials to the projector augmented-wave method, *Physical Review B* **59** (1999) 1758.
- [19] M. Shishkin and G. Kresse, Implementation and performance of the frequency-dependent *GW* method within the PAW framework, *Physical Review B* **74** (2006) 035101.
- [20] H.J. Monkhorst and J.D. Pack, Special points for Brillouin-zone integrations, *Physical Review B* **13** (1976) 5188.
- [21] Density functional theory study of the relative stability of the iron disulfide polymorphs pyrite and marcasite, *Physical Review B* **81** (2010) 094106.
- [22] R. Sun, M.K.Y. Chan, and G. Ceder, First-principles electronic structure and relative stability of pyrite and marcasite: Implications for photovoltaic performance, *Physical Review B* **83** (2011) 235311.
- [23] V.K. Gudelli, V. Kanchana, S. Appalakondaiah, G. Vaitheeswaran, M.C. Valsakumar, Phase Stability and Thermoelectric Properties of the Mineral FeS₂: An Ab Initio Study, *The Journal of Physical Chemistry C* **117** (2013) 21120–21131.